

28.5 Post-selection and debiased inference

This section turns to confidence intervals and tests for a single coefficient β_j when $p > n$. Two tempting approaches fail: least squares intervals for the model the lasso selected, as if it had been fixed in advance, and intervals around the lasso estimate itself, whose shrinkage bias is at least as large as its standard error. The fix is a one-step correction of the lasso, which turns out to be the Frisch–Waugh–Lovell theorem with a lasso in place of a projection.

28.5.1 Why intervals after selection fail

The least squares interval of Theorem 12.1.1 has coverage $1 - \alpha$ for a model fixed before the data are seen. After selection, the interval for β_j is computed only in the samples where x_j was selected, which are those in which its estimated effect happened to be large. Conditional on selection the estimate is biased away from zero, and the interval is centred in the wrong place. This is the winner’s curse of Chapter 29 (Proposition 29.5.1), and the multiplicity problem of Chapter 13 in another form. With the lasso and $p > n$ it is severe, because the selection is among hundreds or thousands of candidates.

Leeb and Pötscher (2006) proved that this cannot be repaired by estimating the distribution of the post-selection estimator, which depends on the unknown coefficients in a way that no estimator can track uniformly. One can change the target (the selected model’s coefficients, conditional on the selection) or the estimator. This section changes the estimator.

The lasso estimate itself is no better as a centre for an interval. On the event of Theorem 28.4.2, Eq. 28.4.1 shows that it equals least squares on S minus $\lambda \hat{\Sigma}_{SS}^{-1} \text{sign}(\beta_S)$. The shift is of order $\lambda \asymp \sigma \sqrt{\log p/n}$, which is larger than the standard error $\sigma/\sqrt{n}$ of the oracle estimate by the factor $\sqrt{\log p}$. An interval of the usual width centred at the lasso estimate misses the truth more and more often as p grows.

28.5.2 The debiased lasso

Write $\hat{\Sigma} = \mathbf{X}^\top \mathbf{X}/n$, and let $\hat{\Theta}$ be a $p \times p$ matrix computed from $\mathbf{X}$ alone, meant to act like an inverse of $\hat{\Sigma}$ (which does not exist when $p > n$). Its rows are $\hat{\theta}_1^\top, \dots, \hat{\theta}_p^\top$.

Definition 28.5.1 — Debiased lasso

Given a lasso estimate $\hat{\beta}_\lambda$ and $\hat{\Theta}$, the **debiased** (or **desparsified**) lasso is

$$\hat{\mathbf{b}} = \hat{\beta}_\lambda + \frac{1}{n} \hat{\Theta} \mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\beta}_\lambda). \quad (28.5.1)$$

Its j th entry uses only the row $\hat{\theta}_j$. The quantity $\mu_j = \|\hat{\Sigma} \hat{\theta}_j - \mathbf{e}_j\|_\infty$ measures how far $\hat{\theta}_j$ is from being the j th row of an inverse of $\hat{\Sigma}$.

The correction adds to the lasso a multiple of the residual correlations $\mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\beta}_\lambda)/n$. If $\hat{\Sigma}$ were invertible and $\hat{\Theta} = \hat{\Sigma}^{-1}$, $\hat{\mathbf{b}}$ would be the least squares estimate (Exercise (A1)).

Theorem 28.5.1 — Debiased lasso

In the setting of [Section 28.3](#), let $\hat{\Theta}$ depend on $\mathbf{X}$ only, and let $\hat{\mathbf{b}}$ be the debiased lasso.

- (a) (*Decomposition.*) $\sqrt{n}(\hat{\mathbf{b}} - \boldsymbol{\beta}) = \mathbf{w} + \mathbf{r}$, where $\mathbf{w} = \hat{\Theta}\mathbf{X}^\top \boldsymbol{\varepsilon}/\sqrt{n} \sim N_p(\mathbf{0}, \sigma^2 \hat{\Theta} \hat{\Sigma} \hat{\Theta}^\top)$ exactly, and $\mathbf{r} = -\sqrt{n}(\hat{\Theta} \hat{\Sigma} - \mathbf{I})(\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta})$.
- (b) (*Remainder.*) $|r_j| \leq \sqrt{n} \mu_j \|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|_1$. If $\lambda \geq 2\lambda_0$, $\phi^2(S) > 0$ and $\mathcal{T}$ occurs, then $|r_j| \leq 4\sqrt{n} \mu_j \lambda s / \phi^2(S)$.
- (c) (*Variance.*) If $\mu_j < 1$, then $\omega_j^2 = \hat{\boldsymbol{\theta}}_j^\top \hat{\Sigma} \hat{\boldsymbol{\theta}}_j \geq (1 - \mu_j)^2$.
- (d) (*Asymptotic normality.*) Consider a sequence of such problems with $n \rightarrow \infty$, in which p , s , $\mathbf{X}$, $\boldsymbol{\beta}$ and $\hat{\Theta}$ may change with n and σ is fixed. Suppose $\lambda = 2\lambda_0$ with $\delta = \delta_n \rightarrow 0$ and $\log(1/\delta_n) = O(\log p)$; $\phi^2(S) \geq \phi_0^2 > 0$; $\mu_j \leq C\sqrt{\log p/n}$ for a constant C ; $s \log p / \sqrt{n} \rightarrow 0$; and $\hat{\sigma}/\sigma \rightarrow 1$ in probability. Then

$$T_j = \frac{\sqrt{n}(\hat{b}_j - \beta_j)}{\hat{\sigma} \omega_j} \rightarrow N(0, 1) \quad \text{in distribution,} \quad (28.5.2)$$

and the interval $\hat{b}_j \pm z_{\alpha/2} \hat{\sigma} \omega_j / \sqrt{n}$ has coverage tending to $1 - \alpha$.

Proof.

- (a) Substitute $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ in [Eq. 28.5.1](#) and write $\boldsymbol{\Delta} = \hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}$:

$$\hat{\mathbf{b}} - \boldsymbol{\beta} = \boldsymbol{\Delta} + \frac{1}{n} \hat{\Theta} \mathbf{X}^\top (\boldsymbol{\varepsilon} - \mathbf{X} \boldsymbol{\Delta}) = \frac{1}{n} \hat{\Theta} \mathbf{X}^\top \boldsymbol{\varepsilon} - (\hat{\Theta} \hat{\Sigma} - \mathbf{I}) \boldsymbol{\Delta}.$$

Multiply by $\sqrt{n}$. Since $\hat{\Theta}$ is a fixed matrix, $\mathbf{w}$ is a linear function of $\boldsymbol{\varepsilon}$, hence normal with mean zero and covariance $\hat{\Theta} \mathbf{X}^\top (\sigma^2 \mathbf{I}) \mathbf{X} \hat{\Theta}^\top / n = \sigma^2 \hat{\Theta} \hat{\Sigma} \hat{\Theta}^\top$ (Theorem 3.3.1).

- (b) The j th row of $\hat{\Theta} \hat{\Sigma} - \mathbf{I}$ is $(\hat{\Sigma} \hat{\boldsymbol{\theta}}_j - \mathbf{e}_j)^\top$, because $\hat{\Sigma}$ is symmetric. By Hölder's inequality $|r_j| \leq \sqrt{n} \mu_j \|\boldsymbol{\Delta}\|_1$. The second bound follows from Theorem 28.3.4(a).
- (c) The columns of $\mathbf{X}$ have $\|\mathbf{x}_j\|^2 = n$, so $\mathbf{e}_j^\top \hat{\Sigma} \mathbf{e}_j = 1$. The j th entry of $\hat{\Sigma} \hat{\boldsymbol{\theta}}_j$ differs from 1 by at most μ_j , so $\mathbf{e}_j^\top \hat{\Sigma} \hat{\boldsymbol{\theta}}_j \geq 1 - \mu_j > 0$. The Cauchy–Schwarz inequality for the nonnegative definite form $\hat{\Sigma}$ (Proposition 1.1.1 applied to $\hat{\Sigma}^{1/2} \mathbf{e}_j$ and $\hat{\Sigma}^{1/2} \hat{\boldsymbol{\theta}}_j$) gives $(1 - \mu_j)^2 \leq (\mathbf{e}_j^\top \hat{\Sigma} \hat{\boldsymbol{\theta}}_j)^2 \leq (\mathbf{e}_j^\top \hat{\Sigma} \mathbf{e}_j) \omega_j^2 = \omega_j^2$.
- (d) Write $T_j = (\sigma/\hat{\sigma})\{w_j/(\sigma \omega_j) + r_j/(\sigma \omega_j)\}$. By (a), $w_j/(\sigma \omega_j)$ has exactly the $N(0, 1)$ law. For the remainder, $\log(2p/\delta_n) = \log 2 + \log p + \log(1/\delta_n) \leq C' \log p$ for a constant C' and large n , so $\lambda = 2\lambda_0 \leq 2\sigma \sqrt{2C' \log p/n}$. On $\mathcal{T}$, which has probability at least $1 - \delta_n \rightarrow 1$, part (b) gives

$$\begin{aligned} |r_j| &\leq 4\sqrt{n} \cdot C \sqrt{\frac{\log p}{n}} \cdot 2\sigma \sqrt{\frac{2C' \log p}{n}} \cdot \frac{s}{\phi_0^2} \\ &= \frac{8\sqrt{2C'} C \sigma}{\phi_0^2} \cdot \frac{s \log p}{\sqrt{n}} \rightarrow 0. \end{aligned}$$

By (c) and $\mu_j \rightarrow 0$, $\omega_j \geq \frac{1}{2}$ for large n . Hence $r_j/(\sigma \omega_j) \rightarrow 0$ in probability, and Slutsky's lemma (with $\sigma/\hat{\sigma} \rightarrow 1$) gives [Eq. 28.5.2](#). The coverage statement is $\Pr(|T_j| \leq z_{\alpha/2}) \rightarrow 1 - \alpha$.

■

The theorem separates what the design must deliver (a $\hat{\Theta}$ with small μ_j , and $\phi^2(S)$ bounded below) from what the lasso delivers (an ℓ_1 error of order $s\sqrt{\log p/n}$). The condition $s \log p/\sqrt{n} \rightarrow 0$ is stronger than the $s \log p/n \rightarrow 0$ that suffices for prediction: the remainder must be small compared with the standard error $1/\sqrt{n}$.

28.5.3 Constructing the approximate inverse

The remaining question is how to find a $\hat{\Theta}$ with small μ_j when $\hat{\Sigma}$ is singular. The *node-wise lasso* regresses each column on all the others.

Lemma 28.5.2 — Node-wise lasso

For a fixed j , let $\hat{\gamma}_j \in \mathbb{R}^{p-1}$ be a lasso estimate, with penalty λ_j , for the regression of $\mathbf{x}_j$ on the other columns $\mathbf{X}_{-j}$. Put $\hat{\mathbf{z}}_j = \mathbf{x}_j - \mathbf{X}_{-j}\hat{\gamma}_j$ and $\hat{\tau}_j^2 = \mathbf{x}_j^\top \hat{\mathbf{z}}_j/n$, and let $\hat{\theta}_j$ have entry $1/\hat{\tau}_j^2$ in position j and $-\hat{\gamma}_j/\hat{\tau}_j^2$ elsewhere. Then:

- (a) $\hat{\tau}_j^2 = \|\hat{\mathbf{z}}_j\|^2/n + \lambda_j \|\hat{\gamma}_j\|_1 \geq 0$;
- (b) if $\hat{\tau}_j^2 > 0$, then $(\hat{\Sigma}\hat{\theta}_j)_j = 1$ and $\mu_j = \|\hat{\Sigma}\hat{\theta}_j - \mathbf{e}_j\|_\infty \leq \lambda_j/\hat{\tau}_j^2$;
- (c) the j th debiased estimate is

$$\hat{b}_j = \frac{\hat{\mathbf{z}}_j^\top (\mathbf{Y} - \mathbf{X}_{-j}\hat{\beta}_{\lambda,-j})}{\hat{\mathbf{z}}_j^\top \mathbf{x}_j}. \quad (28.5.3)$$

Proof. The optimality conditions Eq. 28.3.2 for the node-wise regression say that

$$\mathbf{X}_{-j}^\top \hat{\mathbf{z}}_j/n = \lambda_j \hat{\mathbf{k}},$$

where $\hat{\mathbf{k}}$ has entries of size at most 1 and equal to $\text{sign}(\hat{\gamma}_{j,k})$ where $\hat{\gamma}_{j,k} \neq 0$. Hence $\hat{\gamma}_j^\top \mathbf{X}_{-j}^\top \hat{\mathbf{z}}_j/n = \lambda_j \|\hat{\gamma}_j\|_1$, and since $\mathbf{x}_j = \hat{\mathbf{z}}_j + \mathbf{X}_{-j}\hat{\gamma}_j$,

$$\hat{\tau}_j^2 = \frac{\mathbf{x}_j^\top \hat{\mathbf{z}}_j}{n} = \frac{\|\hat{\mathbf{z}}_j\|^2}{n} + \lambda_j \|\hat{\gamma}_j\|_1,$$

which is (a). For (b), $\mathbf{X}\hat{\theta}_j = (\mathbf{x}_j - \mathbf{X}_{-j}\hat{\gamma}_j)/\hat{\tau}_j^2 = \hat{\mathbf{z}}_j/\hat{\tau}_j^2$, so the k th entry of $\hat{\Sigma}\hat{\theta}_j$ is $\mathbf{x}_k^\top \hat{\mathbf{z}}_j/(n\hat{\tau}_j^2)$. For $k = j$ this is 1 by the definition of $\hat{\tau}_j^2$. For $k \neq j$ it is $\lambda_j \hat{k}_k/\hat{\tau}_j^2$, of size at most $\lambda_j/\hat{\tau}_j^2$. For (c), $\hat{b}_j$ is $\hat{\beta}_{\lambda,j} + \hat{\mathbf{z}}_j^\top (\mathbf{Y} - \mathbf{X}\hat{\beta}_\lambda)/(n\hat{\tau}_j^2)$. Put $n\hat{\tau}_j^2 = \hat{\mathbf{z}}_j^\top \mathbf{x}_j$ and $\mathbf{X}\hat{\beta}_\lambda = \mathbf{x}_j\hat{\beta}_{\lambda,j} + \mathbf{X}_{-j}\hat{\beta}_{\lambda,-j}$; the terms in $\hat{\beta}_{\lambda,j}$ cancel. ■

Part (c) shows what the debiased lasso is. By the Frisch–Waugh–Lovell theorem (Theorem 6.6.2), the least squares coefficient of $\mathbf{x}_j$ comes from regressing $\mathbf{Y}$ on the part of $\mathbf{x}_j$ orthogonal to the other columns, which is zero when $p > n$. The node-wise lasso replaces it by a lasso residual $\hat{\mathbf{z}}_j$, nearly orthogonal to every other column by (b), and removes the other regressors' effect approximately through $\mathbf{X}_{-j}\hat{\beta}_{\lambda,-j}$. Javanmard and Montanari (2014) instead choose $\hat{\theta}_j$ to minimize $\hat{\theta}^\top \hat{\Sigma} \hat{\theta}$ subject to $\|\hat{\Sigma} \hat{\theta} - \mathbf{e}_j\|_\infty \leq \mu$.

Remark (Sketch: when the assumptions of part (d) hold). This is a sketch; the full arguments are in van de Geer, Bühlmann, Ritov and Dezeure (2014) and Javanmard and Montanari (2014). Suppose the rows of $\mathbf{X}$ are independent $N_p(\mathbf{0}, \Sigma)$, the eigenvalues of Σ are bounded away from 0 and ∞ , and row j of Σ^{-1} has at most s_j nonzero entries. The population regression of x_j on the others then has s_j nonzero coefficients and residual variance $\tau_j^2 = 1/(\Sigma^{-1})_{jj}$. By the lemma $\mu_j \leq \lambda_j/\hat{\tau}_j^2$ *deterministically*, so with $\lambda_j \asymp \sqrt{\log p/n}$ it remains to show $\hat{\tau}_j^2 \approx \tau_j^2$. This follows from Section 28.3 applied to the node-wise regression, with the random-design bound of Section 28.2, when $s_j \log p/n \rightarrow 0$; the scaled lasso (Sun and Zhang 2012) supplies $\hat{\sigma}$. Then $\omega_j^2 \rightarrow (\Sigma^{-1})_{jj}$, which van de Geer and coauthors show to be the smallest asymptotic variance of a regular estimator.

Warning. The theorem covers one prespecified coefficient, or a fixed finite set. Coefficients chosen *because* they look large bring back the selection problem, and simultaneous intervals need a multiplicity correction such as Bonferroni (Theorem 13.4.1). With moderate n and a truth that is not very sparse, the remainder r_j can be comparable to the noise, and coverage suffers.

28.5.4 A simulation

Example. (Coverage of three intervals)

Take $n = 200$, $p = 400$, a fixed Gaussian design with correlations $0.5^{|j-k|}$, $\sigma = 1$, and $\beta_1 = 1$, $\beta_6 = 0.15$, $\beta_{11} = -1$, $\beta_{16} = 0.15$, all other coefficients zero. The targets are a strong coefficient β_1 , a weak one β_6 , and $\beta_2 = 0$, whose regressor is correlated 0.5 with x_1 . In each of 1000 data sets we fit the lasso with $\lambda = \sigma\sqrt{2\log p/n} = 0.245$, below $2\lambda_0$ for the reason seen in Example (How tight are the bounds?), and compute three 95% intervals:

- the **naïve** interval: least squares on the selected variables and its t interval, reported only when the target was selected;
- the **debiased** interval $\hat{b}_j \pm 1.96 \hat{\sigma}\omega_j/\sqrt{n}$, with node-wise lasso rows ($\lambda_j = 0.024$) and $\hat{\sigma}^2$ the lasso residual sum of squares divided by n minus the number of selected variables;
- the **oracle** t interval from least squares on the true support plus the target.

The results are in Figure 28.5.1. For the null coefficient β_2 , the lasso selected x_2 in a fraction 0.069 of the data sets, and in those the naïve interval covered zero only in a fraction 0.174. For the weak coefficient, selected in a fraction 0.329, the naïve coverage was 0.903, and for the strong one 0.915. The debiased intervals covered β_1 , β_6 and β_2 in fractions 0.960, 0.946 and 0.940 respectively, close to the oracle's 0.943, 0.956 and 0.958 for the same three coefficients.

On the strong coefficient the lasso's average error is -0.210 , close to $-\lambda$, and the debiased estimate's is 0.014. The standardized estimates T_1 have mean 0.14 and standard deviation 0.95 (panel (a)). Two costs remain.

The debiased intervals are wider than the oracle's (0.393 against 0.288 for β_1): part of this is the price of not knowing the support, and part is tuning. The node-wise penalty $\lambda_j = 0.024$, a tenth of $\sqrt{2\log p/n}$, keeps the bound $\mu_j \leq \lambda_j/\hat{\tau}_j^2$ of Lemma 28.5.2, and

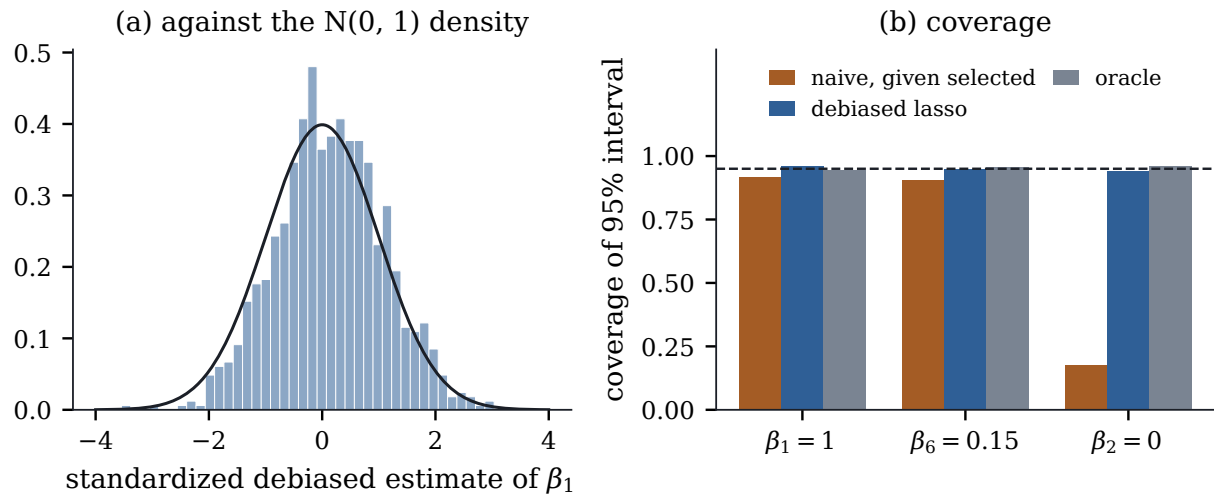


Figure 28.5.1: Inference for single coefficients with $n = 200$, $p = 400$, 1000 data sets. (a) Histogram of the standardized debiased estimates of β_1 and the $N(0, 1)$ density. (b) Coverage of nominal 95% intervals: naive least squares after lasso selection (computed only when the target is selected), the debiased lasso, and the oracle.

so the remainder, small. But it lets the node-wise fits explain much of each column: $\hat{\tau}_j^2$ (0.28, 0.25 and 0.27 for β_1 , β_6 and β_2) is well below the population value $1/(\Sigma^{-1})_{jj}$, which is $\frac{3}{4}$ for $j = 1$ and $\frac{3}{5}$ otherwise, and that allows a larger ω_j (Exercise (B2)).

The second cost is a remainder bias: 0.049 for β_2 and 0.046 for the weak β_6 , about a third of its value. This is the term $r_j/\sqrt{n}$ of Theorem 28.5.1, at most $\mu_j\|\hat{\beta}_\lambda - \beta\|_1$, with μ_j between 0.088 and 0.099 and an average lasso ℓ_1 error of 0.74. At $n = 200$ that product is not negligible.

```
import numpy as np
from scipy import stats

def soft(z, t):
    """Soft thresholding  $S(z, t) = \text{sign}(z) \max(|z| - t, 0)$ ."""
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)

def lasso_cd(X, Y, lam, B=None, tol=1e-9, max_sweeps=2000):
    """Minimize  $\|y - Xb\|_2^2 / (2n) + \text{lam} \|b\|_1$  for every column  $y$  of  $Y$ , by
    coordinate descent."""
    n, p = X.shape
    Y = Y.reshape(n, -1)
    B = np.zeros((p, Y.shape[1])) if B is None else B.copy()
    R = Y - X @ B # residuals, one column per
    response
    scale = np.sum(X ** 2, axis=0) / n
    active = np.arange(p)
    for sweep in range(max_sweeps):
        change = 0.0
        for j in active:
            z = X[:, j] @ R / n + scale[j] * B[j]
            new = soft(z, lam) / scale[j]
            d = new - B[j]
            if np.any(d != 0):
                R -= np.outer(X[:, j], d)
                B[j] = new
```

```

        change = max(change, np.max(np.abs(d)))
    if change < tol:
        if len(active) == p:                                # converged over all
coordinates
            break
        active = np.arange(p)                                # check every coordinate once
more
    else:
        active = np.flatnonzero(np.any(B != 0, axis=1)) if sweep % 5 else np.
arange(p)
    return B

def nodewise_row(X, j, lam_j):
    """Row theta_j of Theta from the lasso of x_j on the other columns, and tau_j^2.
    """
    n, p = X.shape
    others = np.delete(np.arange(p), j)
    gamma = lasso_cd(X[:, others], X[:, j], lam_j)[: , 0]
    z = X[:, j] - X[:, others] @ gamma                    # lasso residual of x_j on the
rest
    tau2 = X[:, j] @ z / n
    theta = np.zeros(p)
    theta[j] = 1.0
    theta[others] = -gamma
    return theta / tau2, tau2

def debiased_interval(X, Y, B, j, theta, level=0.95):
    """Debiased estimate of beta_j and its normal-theory interval, one per column of
    Y. """
    n = X.shape[0]
    R = Y - X @ B
    b = B[j] + theta @ X.T @ R / n                        # one-step correction
    sigma_hat = np.sqrt(np.sum(R ** 2, axis=0) / (n - np.sum(B != 0, axis=0)))
    G = X.T @ X / n
    se = sigma_hat * np.sqrt(theta @ G @ theta / n)
    z = stats.norm.ppf((1 + level) / 2)
    return b, b - z * se, b + z * se

def toeplitz_design(rng, n, p, rho):
    idx = np.arange(p)
    L = np.linalg.cholesky(rho ** np.abs(idx[:, None] - idx[None, :]))
    X = rng.normal(size=(n, p)) @ L.T
    return X / np.sqrt(np.sum(X ** 2, axis=0) / n)

rng = np.random.default_rng(2850)
n, p, sigma = 200, 400, 1.0
X = toeplitz_design(rng, n, p, rho=0.5)
beta = np.zeros(p)
beta[[0, 5, 10, 15]] = [1.0, 0.15, -1.0, 0.15]
lam = sigma * np.sqrt(2 * np.log(p) / n)
targets = {"strong": 0, "weak": 5, "null": 1}

Ys = (X @ beta)[: , None] + sigma * rng.normal(size=(n, 200))
Bs = lasso_cd(X, Ys, lam)
for name, j in targets.items():
    theta, tau2 = nodewise_row(X, j, 0.1 * np.sqrt(2 * np.log(p) / n))
    b, lo, hi = debiased_interval(X, Ys, Bs, j, theta)
    print(f"{name:6s}: lasso bias {Bs[j].mean() - beta[j]:+.3f}, debiased bias {b.

```

```
mean() - beta[j]:+.3f},"
f" coverage {np.mean((lo <= beta[j]) & (beta[j] <= hi)):.3f}"
```

[Run this code online](#)

28.5.5 Other routes to valid inference

Other approaches answer different questions. **Sample splitting** (Wasserman and Roeder 2009) selects on one half of the data and fits on the other ([Exercise B1](#)). **Selective inference** (Lee, Sun, Sun and Taylor 2016) conditions on the lasso having chosen the reported model, a union of polyhedra in $\mathbf{Y}$. **Simultaneous inference over all models** (Berk, Brown, Buja, Zhang and Zhao 2013) widens the intervals, as Scheffé's method does. These target the coefficients of the *selected* model, which generally differ from β_j . **Double selection** (Belloni, Chernozhukov and Hansen 2014) targets β_j : it keeps the controls that predict the response or x_j and refits least squares, the idea of [Eq. 28.5.3](#).

28.5.6 Exercises

28.5.6.1 A. Check your understanding

Exercise A1.

Suppose $p < n$, $\mathbf{X}$ has full column rank and $\hat{\Theta} = \hat{\Sigma}^{-1}$. Show that the debiased lasso [Eq. 28.5.1](#) equals the least squares estimate, whatever $\hat{\beta}_\lambda$ is, and that $\mu_j = 0$ and $\mathbf{r} = \mathbf{0}$ in [Theorem 28.5.1](#).

Solution. $\hat{\mathbf{b}} = \hat{\beta}_\lambda + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\beta}_\lambda) = \hat{\beta}_\lambda + \hat{\beta} - \hat{\beta}_\lambda = \hat{\beta}$, where $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$. Then $\hat{\Sigma} \hat{\Theta} = \mathbf{I}$, so every $\mu_j = 0$ and $\mathbf{r} = \mathbf{0}$, and part (a) of the theorem reduces to $\sqrt{n}(\hat{\beta} - \beta) \sim N_p(\mathbf{0}, \sigma^2 \hat{\Sigma}^{-1})$ (Theorem 7.5.1).

Exercise A2.

On the event of Theorem 28.4.2, with $\hat{\Sigma}_{SS} = \mathbf{I}$, show that the interval $\hat{\beta}_{\lambda,j} \pm 1.96 \sigma / \sqrt{n}$ for $j \in S$ covers β_j iff $|\mathbf{x}_j^\top \boldsymbol{\varepsilon} / \sqrt{n} \sigma^{-1} - \sqrt{n} \lambda \text{sign}(\beta_j) / \sigma| \leq 1.96$. If $\lambda = 2\lambda_0$, what happens to its coverage as $p \rightarrow \infty$?

28.5.6.2 B. Practice

Exercise B1.

(*Sample splitting.*) Split the observations into halves I_1 and I_2 chosen independently of the data. Select a set $\hat{S}$ of columns using only the data in I_1 , and suppose $\mathbf{X}_{I_2, \hat{S}}$ has full column rank and $|\hat{S}| < |I_2|$. Show that if $\hat{S} \supseteq S$, then conditionally on the data in I_1 , the usual t interval for β_j ($j \in \hat{S}$) from the least squares fit of $\mathbf{Y}_{I_2}$ on $\mathbf{X}_{I_2, \hat{S}}$ has exact coverage $1 - \alpha$. What does the fit estimate when $\hat{S}$ misses an active variable?

Solution. The errors in I_2 are independent of those in I_1 , so conditionally on the data in I_1 the set $\hat{S}$ is fixed and $\mathbf{Y}_{I_2} \sim \mathcal{N}(\mathbf{X}_{I_2}\boldsymbol{\beta}, \sigma^2\mathbf{I})$. If $\hat{S} \supseteq S$, then $\mathbf{X}_{I_2}\boldsymbol{\beta} = \mathbf{X}_{I_2, \hat{S}}\boldsymbol{\beta}_{\hat{S}}$, so the fit on $\mathbf{X}_{I_2, \hat{S}}$ is a correctly specified normal linear model with coefficients $\boldsymbol{\beta}_{\hat{S}}$, and the t interval of Theorem 12.1.1 has exact conditional coverage. Averaging over the data in I_1 , coverage is exact on the event $\hat{S} \supseteq S$. If $\hat{S}$ misses an active variable, the fit estimates the projected coefficients $(\mathbf{X}_{I_2, \hat{S}}^\top \mathbf{X}_{I_2, \hat{S}})^{-1} \mathbf{X}_{I_2, \hat{S}}^\top \mathbf{X}_{I_2} \boldsymbol{\beta}$ (Proposition 5.5.7), which differ from $\boldsymbol{\beta}_{\hat{S}}$ when the missed variables are correlated with the selected ones, and the residual mean square overestimates σ^2 .

Exercise B2.

In Lemma 28.5.2, show that $\omega_j^2 = \hat{\boldsymbol{\theta}}_j^\top \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j = \|\hat{\mathbf{z}}_j\|^2 / (n\hat{\tau}_j^4)$, and hence $\omega_j^2 \leq 1/\hat{\tau}_j^2$. Interpret: a small $\hat{\tau}_j^2$ (a column that the others explain well) means a large variance for $\hat{b}_j$. Compare with the variance inflation factor (Definition 26.2.1).

Solution. $\mathbf{X}\hat{\boldsymbol{\theta}}_j = \hat{\mathbf{z}}_j/\hat{\tau}_j^2$, so $\omega_j^2 = \|\mathbf{X}\hat{\boldsymbol{\theta}}_j\|^2/n = \|\hat{\mathbf{z}}_j\|^2/(n\hat{\tau}_j^4)$. By Lemma 28.5.2(a), $\|\hat{\mathbf{z}}_j\|^2/n \leq \hat{\tau}_j^2$, so $\omega_j^2 \leq 1/\hat{\tau}_j^2$. With least squares in place of the lasso ($p < n$, $\lambda_j = 0$), $\hat{\tau}_j^2 = \|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2/n$, and $\omega_j^2 = 1/\hat{\tau}_j^2$ is the variance factor $n/\{S_{jj}(1 - R_j^2)\}$ of Eq. 6.6.2 for a column with $\|\mathbf{x}_j\|^2 = n$ (and an intercept among the other columns). The node-wise lasso residual plays the role of the FWL residual, and $1/\hat{\tau}_j^2$ the role of the variance inflation.

28.5.6.3 C. Going deeper

Exercise C1.

The two quantities in Theorem 28.5.1 pull against each other: μ_j controls the remainder and ω_j the standard error, and a $\hat{\boldsymbol{\theta}}_j$ that makes one small makes the other large. This exercise computes the trade-off exactly.

- Suppose $p < n$ and $\hat{\boldsymbol{\Sigma}}$ is invertible. Show that the only $\boldsymbol{\theta}$ with $\mu_j = 0$, that is with $\hat{\boldsymbol{\Sigma}}\boldsymbol{\theta} = \mathbf{e}_j$, gives $\omega_j^2 = (\hat{\boldsymbol{\Sigma}}^{-1})_{jj}$, the least squares variance factor of Exercise (B2).
- Let $p = 2$, $\hat{\boldsymbol{\Sigma}} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ with $0 < \rho < 1$, and $j = 1$. Substituting $(a, b)^\top = \hat{\boldsymbol{\Sigma}}\boldsymbol{\theta}$, show that $\boldsymbol{\theta}^\top \hat{\boldsymbol{\Sigma}}\boldsymbol{\theta} = (a^2 - 2\rho ab + b^2)/(1 - \rho^2)$ while the constraint $\|\hat{\boldsymbol{\Sigma}}\boldsymbol{\theta} - \mathbf{e}_1\|_\infty \leq \mu$ reads $|a - 1| \leq \mu$ and $|b| \leq \mu$.
- Show that for $0 < \mu \leq \rho/(1 + \rho)$ the minimum is at $(a, b) = (1 - \mu, \mu)$, with value

$$\omega_1^2(\mu) = \frac{1 - 2(1 + \rho)\mu(1 - \mu)}{1 - \rho^2},$$

so that $\omega_1^2(0) = (\hat{\boldsymbol{\Sigma}}^{-1})_{11}$ and $\frac{d}{d\mu}\omega_1^2(0) = -2/(1 - \rho)$.

- Read the result against Theorem 28.5.1(b) and (c), and against the tuning reported in
- (a)

Solution.

- (a) $\hat{\Sigma}\theta = \mathbf{e}_j$ has the unique solution $\theta = \hat{\Sigma}^{-1}\mathbf{e}_j$, and $\theta^\top \hat{\Sigma}\theta = \mathbf{e}_j^\top \hat{\Sigma}^{-1}\mathbf{e}_j = (\hat{\Sigma}^{-1})_{jj}$. By [Exercise \(A1\)](#) the debiased estimate is then least squares, with exactly that variance factor.
- (b) $\theta = \hat{\Sigma}^{-1}(a, b)^\top$, so $\theta^\top \hat{\Sigma}\theta = (a, b)\hat{\Sigma}^{-1}(a, b)^\top$, and $\hat{\Sigma}^{-1} = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$ gives the stated form. The constraint is a constraint on (a, b) directly.
- (c) Write $f(a, b) = (a^2 - 2\rho ab + b^2)/(1 - \rho^2)$ on the box $[1 - \mu, 1 + \mu] \times [-\mu, \mu]$. Then $\partial f/\partial a = 2(a - \rho b)/(1 - \rho^2)$, and on the box $a - \rho b \geq (1 - \mu) - \rho\mu > 0$ because $\mu \leq \rho/(1 + \rho) < 1/(1 + \rho)$; so f increases in a and $a = 1 - \mu$. Next $\partial f/\partial b = 2(b - \rho a)/(1 - \rho^2)$ vanishes at $b = \rho(1 - \mu)$, which exceeds μ exactly when $\mu < \rho/(1 + \rho)$; so f decreases in b throughout $[-\mu, \mu]$ and $b = \mu$. Substituting, $(1 - \mu)^2 - 2\rho(1 - \mu)\mu + \mu^2 = 1 - 2(1 + \rho)\mu + 2(1 + \rho)\mu^2$, which is the stated numerator. Setting $\mu = 0$ gives $1/(1 - \rho^2)$, and differentiating at $\mu = 0$ gives $-2(1 + \rho)/(1 - \rho^2) = -2/(1 - \rho)$.
- (d) A relaxation of the constraint from $\mu = 0$ to a small μ lowers the variance factor at the linear rate $2/(1 - \rho)$, fast when the columns are correlated, while by (b) of the theorem the remainder bound $\sqrt{n}\mu_j\|\hat{\beta}_\lambda - \beta\|_1$ grows linearly in μ at the same time. The theorem's condition $\mu_j \leq C\sqrt{\log p/n}$ is a choice of a point on this curve: it is what makes the remainder vanish beside $1/\sqrt{n}$, at the price of the variance the exact inverse would have delivered. Part (c) of the theorem gives the floor $\omega_j^2 \geq (1 - \mu_j)^2$, so the variance cannot be driven below about one however μ is chosen. The simulation of 1 sits at the other end: its node-wise penalty is a tenth of $\sqrt{2\log p/n}$, which keeps μ_j near 0.09 and the remainder small, and pays for it with a $\hat{\tau}_j^2$ far below its population value and hence intervals wider than the oracle's.

Exercise C2.

Why one correction and not two? Write $T(\mathbf{b}) = \mathbf{b} + \hat{\Theta}\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\mathbf{b})/n$, so that $\hat{\mathbf{b}} = T(\hat{\beta}_\lambda)$, and put $\mathbf{K} = \hat{\Theta}\hat{\Sigma} - \mathbf{I}$.

- (a) Show that $T(\mathbf{b}) - \beta = -\mathbf{K}(\mathbf{b} - \beta) + \mathbf{w}/\sqrt{n}$ with $\mathbf{w}$ as in [Theorem 28.5.1\(a\)](#), and deduce that the k -fold iterate satisfies

$$\sqrt{n}(T^k(\hat{\beta}_\lambda) - \beta) = (-\mathbf{K})^k \sqrt{n} \Delta + \sum_{i=0}^{k-1} (-\mathbf{K})^i \mathbf{w}, \quad \Delta = \hat{\beta}_\lambda - \beta.$$

- (b) With $\mu = \max_k \mu_k$, show that $\|\mathbf{K}\Delta\|_1 \leq p\mu\|\Delta\|_1$, and hence that the bound of [Theorem 28.5.1\(b\)](#) on the j th remainder is multiplied by $p\mu$ for each extra iteration.
- (c) Show that under the conditions of [Theorem 28.5.1\(d\)](#) with $p > n$ one has $p\mu \geq \sqrt{n\log p}C^{-1} \rightarrow \infty$ up to the constant, so that the guarantee is lost after one extra step, however small μ is.
- (d) Show that when $p < n$ and $\hat{\Theta} = \hat{\Sigma}^{-1}$, $\mathbf{K} = \mathbf{0}$ and every iterate equals the least squares estimate, so nothing of this happens in low dimension.

Solution.

- (a) Substituting $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{Y} - \mathbf{X}\mathbf{b} = \mathbf{X}(\boldsymbol{\beta} - \mathbf{b}) + \boldsymbol{\varepsilon}$, so

$$T(\mathbf{b}) - \boldsymbol{\beta} = (\mathbf{b} - \boldsymbol{\beta}) - \hat{\boldsymbol{\Theta}}\hat{\boldsymbol{\Sigma}}(\mathbf{b} - \boldsymbol{\beta}) + \hat{\boldsymbol{\Theta}}\mathbf{X}^\top \boldsymbol{\varepsilon}/n = -\mathbf{K}(\mathbf{b} - \boldsymbol{\beta}) + \mathbf{w}/\sqrt{n},$$

with $\mathbf{w} = \hat{\boldsymbol{\Theta}}\mathbf{X}^\top \boldsymbol{\varepsilon}/\sqrt{n}$. The map is affine with linear part $-\mathbf{K}$; iterating it k times from $\hat{\boldsymbol{\beta}}_\lambda$ and multiplying by $\sqrt{n}$ gives the display. At $k = 1$ it is [Theorem 28.5.1\(a\)](#).

- (b) The k th row of $\mathbf{K}$ is $(\hat{\boldsymbol{\Sigma}}\hat{\boldsymbol{\theta}}_k - \mathbf{e}_k)^\top$, so $|(\mathbf{K}\boldsymbol{\Delta})_k| \leq \mu_k\|\boldsymbol{\Delta}\|_1 \leq \mu\|\boldsymbol{\Delta}\|_1$ by Hölder, and summing over the p coordinates, $\|\mathbf{K}\boldsymbol{\Delta}\|_1 \leq p\mu\|\boldsymbol{\Delta}\|_1$. Applying the j th-row bound once more, $|\{(\mathbf{K})^2\boldsymbol{\Delta}\}_j| \leq \mu_j\|\mathbf{K}\boldsymbol{\Delta}\|_1 \leq p\mu \cdot \mu_j\|\boldsymbol{\Delta}\|_1$, and by induction each further step costs a factor $p\mu$.
- (c) The one-step remainder is controlled by taking μ of the order $\sqrt{\log p/n}$, which is what the condition $\mu_j \leq C\sqrt{\log p/n}$ asks for and what [Lemma 28.5.2](#) delivers. At that order, and with $p > n$,

$$p\mu \asymp p\sqrt{\frac{\log p}{n}} \geq n\sqrt{\frac{\log p}{n}} = \sqrt{n \log p} \rightarrow \infty.$$

The ℓ_1 norm of the error is spread over p coordinates, and each application of $\mathbf{K}$ trades a factor μ against a factor p ; with p larger than n , p wins.

- (d) Then $\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{\Sigma}} = \mathbf{I}$, so $\mathbf{K} = \mathbf{0}$ and $T(\mathbf{b}) = \boldsymbol{\beta} + \mathbf{w}/\sqrt{n} = \hat{\boldsymbol{\beta}}$ for every $\mathbf{b}$, by [Exercise \(A1\)](#). The iteration converges in one step because the correction is exact.

The moral is that [Eq. 28.5.1](#) is not the first step of a convergent scheme. It is the one step whose error the design can be made to control, and the noise term of (a) also changes as soon as $k > 1$, since $\sum_i (-\mathbf{K})^i \mathbf{w}$ no longer has the covariance $\sigma^2 \omega_j^2$ that the interval uses.